

Kafka Consumer Timeout SOP

Objective

Resolve Kafka consumer timeout issues that cause delayed processing, retries, or unstable consumer group behavior.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T047
Issue	Kafka consumer timeout
Category	Streaming
Service	Kafka
Severity	Medium
Status	Open
Description	Consumer timing out frequently
Expected Resolution	Adjust timeout configurations

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm timeout errors and collect consumer group, topic, and service details.
2. Identify whether timeout is during poll, commit, fetch, or processing.
3. Review network latency between consumer and broker.
4. Check consumer configuration for request.timeout.ms, session.timeout.ms, max.poll.interval.ms, and fetch settings.
5. Validate broker health and topic partition leaders.
6. Inspect downstream dependency latency that may block consumer processing.
7. Tune timeout settings only after confirming actual bottleneck.
8. Monitor timeout rate, lag, and rebalance count after remediation.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Consumer timeout	Broker slow or unreachable	Validate broker and network
Poll timeout	Long processing	Optimize processing or tune max.poll.interval.ms
Commit timeout	Coordinator issue	Check consumer group coordinator and broker health
Fetch timeout	Large messages or slow broker	Tune fetch settings and broker resources

Best Practices

1. Track consumer timeout rate separately from lag.
2. Keep downstream calls outside critical poll loop when possible.
3. Use sensible timeout settings per workload.
4. Review timeouts after Kafka client upgrades.

References

[Apache Kafka Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [Kafka CLI Runbook](#)